

Next word Prediction using n-gram model

Aim: To write a python program to train bi-gram model for a given corpus of text to predict the next probable word given the previous two words of a sentence.

Program:

```
import nltk
from nltk.util import ngrams
from collections import defaultdict

# Step 1: Prepare the corpus (tokenize sentences)
corpus = [
    "The weather is beautiful today",
    "I am learning natural language processing",
    "I am studing ML",
    "Machine learning is fascinating",
    "Language models are powerful tools",
    "I love coding in C++"
    "I love programming in Python"
    "I love todays weather"
]

# Tokenize the corpus (split sentences into words)
tokenized_corpus = [nltk.word_tokenize(sentence.lower()) for sentence in corpus]

# Step 2: Train the bigram model
def train_bigram_model(corpus):
    model = defaultdict(lambda: defaultdict(int)) # Bigram model
    for sentence in corpus:
        # Generate bigrams for each sentence
        bigrams = ngrams(sentence, 2)
        for w1, w2 in bigrams:
            model[w1][w2] += 1 # Increment the count for each bigram
    return model

# Train the bigram model
bigram_model = train_bigram_model(tokenized_corpus)

# Step 3: Calculate word probabilities (bigram probabilities)
def calculate_word_probabilities(model):
    probabilities = defaultdict(lambda: defaultdict(float))
    for w1 in model:
        total_count = float(sum(model[w1].values()))
```

```
    for w2 in model[w1]:
        probabilities[w1][w2] = model[w1][w2] / total_count # Probability of w2 after w1
    return probabilities
```

Calculate word probabilities

```
word_probabilities = calculate_word_probabilities(bigram_model)
```

Step 4: Predict the next word given a context (bigram prediction)

```
def predict_next_word(model, context):
    if context[-1] in model:
        next_word_probs = model[context[-1]]
        next_word = max(next_word_probs, key=next_word_probs.get)
        return next_word
    else:
        return None
```

Take user input for context

```
user_input = input("Enter a sentence or context (e.g., 'The bank'): ")
```

Process the user input (split into words)

```
context = user_input.split()
```

Predict the next word based on the user input

```
predicted_word = predict_next_word(bigram_model, context)
```

Print results

```
print("\nPredicted next word given context '{}':".format(" ".join(context)), predicted_word)
```